

Pattern of Admissions and Needs Assessment for Palliative Care Services Among in-patients in a Tertiary Health Facility in South-Western Nigeria

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Abstract

BACKGROUND

Palliative care development emphasizes education and medication availability. Only 12% of needs for palliative care are met. Palliative care domains and experience of patients and their families are vital in life-limiting conditions. The Lagos University Teaching Hospital (LUTH), possess the National Cancer Centre but currently has no palliative care unit.

AIM

The aim was to examine pattern of admissions and needs assessment for palliative services among patients admitted into LUTH wards.

METHOD

The survey was conducted using a cross-sectional design to recruit 129 consecutive respondents. Together for Short Lives (TfSL) tool was used to categorize respondents' conditions into Palliative and Non-palliative conditions. Information on diagnosis, stage of disease, domains of palliative care, advance care plan, preparation for home care, death and education about the illness was retrieved. Descriptive characteristics of the participants were presented as frequencies and percentages and Chi-squared test was used to determine association between the variables. Data analysis was done using Epi info version 7.2.

RESULTS

80.6% of the respondents had palliative care conditions, 83.7% had family members as their caregiver while 13.2% of the participants had no care-giver and 65.9% had no advance care plan. There was no preparation for home care or death in 72.1% while 70.5% had education about their illness and 68.2% were in the advanced stage of their disease. Participants attending the surgery non-trauma unit (51.6%) were more likely to have advance plans. Adults were more likely to have palliative care conditions (79.8%) compared to children (20.2%), and was statistically significant.

CONCLUSION

Majority of the participants need palliative care services but are unmet and the most predominant condition was cancer. Majority had no advance plan or preparation for home care or death despite having advanced stages of disease. This survey emphasizes the need for symptom management, communication and provision of support.

INTRODUCTION

Palliative care development follows a public health model developed by the WHO that emphasizes policy, education, medication availability, and implementation.¹ Remarkably, approximately 12% of needs for palliative care are met.² Using the WHO Member States regional groupings, over a quarter of adults needing palliative care (26.8%) live in the Western Pacific Region, followed by the Africa region (20.2%), Europe (17.9%) and South East Asia. The figures for the Americas and Eastern Mediterranean regions are 14.1% and 4.0%, respectively. However, the African region has the highest number of adults in need of palliative care per 100,000 adult population, followed by Europe and the Americas regions. The great majority, 76%, of the 53 million adults in need of palliative care live in low- or middle-income countries (LMICs). Most (67%) are made up of individuals aged 50 years and above. However, the highest ratios per 100 000 adult population in need of palliative care are found in the low-income countries. Non-communicable

diseases accounted for 69% of all medical conditions. This could be an indicator for preventable deaths related to inadequate or overwhelmed health care systems in low-income countries. The pattern of cases include lung diseases (5.0%), chronic ischaemic heart diseases (0.8%), non-ischaemic heart diseases (1.8%), cerebrovascular diseases (14.1%), degeneration of the CNS diseases (2.2%), inflammatory diseases of the CNS (0.2%), dementia (12.2%), leukaemia (0.5%), diseases of liver (2.4%), renal failure (1.0%), congenital malformations (0.1%), injury, poisoning, external causes (6.4%), arthrosclerosis (0.2%), musculoskeletal disorders (0.5%), protein energy malnutrition (0.2%), tuberculosis (2.1%), HIV disease (22.2%), malignant neoplasms 28.2%. ²

Medical conditions which may require palliative care encompass numerous and diverse life-limiting conditions and life-threatening conditions that can occur in infants, children, adolescents, and young adults. Due to the sources of data coming mainly from the high-income countries, this list does not contain some of the diagnoses, which are commoner in LMIC such as communicable diseases, extensive burns, and sequelae of severe accidents. The number of diagnostic groups in low income countries needing palliative care has increased including Alzheimer, arteriosclerosis, cerebrovascular disease, chronic ischemic heart disease, congenital malformation, degenerative CNS disease, haemorrhagic fevers, HIV, inflammatory CNS disease, injury, leukaemia, liver disease, low birth weight, premature birth, birth trauma, lung disease, malignant neoplasm, malnutrition, musculoskeletal disorder, non-ischemic heart disease, renal failure, tuberculosis; injury (including poisoning and external causes) has been added and generates 6.4% of the need. The number of symptoms used to identify palliative care need increased from only one (pain) to multiple symptoms including differentiation between mild pain and moderate to severe pain in addition to anxiety/worry, bleeding, confusion/delirium, constipation, dementia, depressed mood, diarrhoea, dry mouth, fatigue, itching, nausea/vomiting, shortness of breath, weakness and wounds. Approximately 40% of patients in low and middle-income countries in need of palliative care are aged 70 years or older, 27% aged 50-69, almost 26% aged 20-49 and only 7% are children. ³

Together for Short Lives (TfSL) came up with a categorization system grouping these conditions into four categories based largely on disease trajectories and outcomes. ⁴ Palliative care domains including Biological, Psychological, Social, Spiritual aspects of care as well as patients' experience including advance plans should be addressed. ⁵ About 18 million people requiring palliative care die in unnecessary pain each year and about 2-4 family members are cared for per patient admitted for palliative care. ² Each year, an estimated 40 million people are in need of palliative care (PC); 78% of them live in low- and middle-income countries. ⁶ Unfortunately, only about 14% of this population currently receive such care. ⁷ It has been estimated that over 25 million people worldwide die with suffering, and more than 6 billion days are lived with suffering each year. ⁸

The Lagos University Teaching Hospital (LUTH), which is one of the largest teaching hospitals in Nigeria, with 950 beds serving the population of Lagos state, houses the National Cancer Centre but currently has no palliative care unit. Thus, there is need to establish a PC unit in LUTH as a form of quality improvement.

The aim of this study was to examine the pattern of admissions and needs assessment for palliative services among patients admitted into the wards of Lagos University Teaching Hospital Idi-Araba in Nigeria. The information gathered will serve as an avenue to expose the gaps in palliative care services and thus, justify the need to establish a palliative care unit in LUTH and in addition, serve as preliminary data for low and middle-income countries examining unmet needs for palliative care.

RESEARCH QUESTION

1. What is the disease pattern of patients admitted into the wards based on using Worldwide need for palliative care for adults based on disease groups (20+ years 2017) and Together for Short Lives (TfSL)?
2. What are the unmet palliative care needs among in-patients?
3. What is the spectrum of distribution of palliative care domains of the patients admitted into the wards including Biological, Psychological, Social and Spiritual domains?

JUSTIFICATION

Palliative care encompasses integrated health services that pay special attention to the specific needs and preferences of individuals, whether physical, psychological, social, or spiritual care, in order to reduce pain, suffering and improve their quality of life and that of their families. Lagos University Teaching Hospital has the largest cancer centre in Nigeria but does not possess palliative care unit or uncertain if palliative care services are delivered to patients admitted into the wards. The aim of the study was to assess the disease pattern, assess the unmet palliative care needs and describe the distribution of palliative care domains of patients admitted into the wards

METHODS

Lagos University Teaching Hospital is a tertiary health facility, located in Surulere local government area of Lagos state Nigeria, with 950 admission beds, forty-six (46) clinical departments, eighteen (18) non-clinical departments. Lagos University Teaching Hospital trains students in clinical and dental sciences, pharmacy and allied courses. LUTH has an advanced cancer treatment which functions under a public-private partnership arrangement between The Nigeria Sovereign Investment Authority (NSIA), and the LUTH. It provides advanced radiotherapy and chemotherapy treatment services. At present, LUTH employs over 2300 staff comprising consultants, resident/medical officers, nurses, pharmacists, physiotherapists, medical laboratory scientists, nutritionists, hospital administrators, engineers and other categories of staff. It is located within two densely populated communities.

Patients recruited were those on admission in wards A, B, D and E blocks in Lagos University Teaching Hospital. Due to the current structural renovation presently on-going in the hospital wards, each ward comprised of mixed clinical specialties due to constraints for bed spaces but admissions into each of the wards are sex-specific for males or females. The maximum bed capacity of the selected wards were A2–30 beds, A3–30 beds, B1–30 beds, B3–30 beds,

D2–32 beds and E6–34 beds. Total number of beds available were 186 out of which 129 participants were recruited for the study (response rate of 69.4%)

Study Design

The design was a descriptive cross-sectional study

Study Population

Information on study objectives was collected from all consenting consecutive patients in the Medical, Surgical, Gynaecological, and Paediatrics wards in 2 week-days (25th – 26th of July, 2022) through interviewer administered proforma. There are usually more admissions in the week-days than week-ends.

The Association for Together for Short Lives (TfSL) came up with a categorization system grouping these conditions into four categories based largely on disease trajectories and outcomes.⁴ Four groups of children with life-limiting and life-threatening conditions: Category 1 TfSL for which curative treatment may be feasible but can fail, where

access to palliative care services may be necessary when treatment fails, irrespective of the duration of that threat to life. These include Cancer, organ failures of heart, liver, kidney, transplant and children on long-term ventilation. Category 2 conditions include when premature death is inevitable; these may involve long periods of intensive disease-directed treatment aimed at prolonging life and allowing participation in normal activities. Children and young people in this category may be significantly disabled but have long periods of relatively good health. These include Cystic fibrosis, Duchenne muscular dystrophy and SMA Type 1 on ventilation, HIV on HAART. Category 3 include progressive conditions without curative treatment options, where treatment is exclusively palliative and may commonly extend over many years. These include Batten disease, mucopolysaccharidoses and other severe metabolic conditions, SMA type 1 without ventilation, HIV no HAART. Category 4 include irreversible but non-progressive conditions causing severe disability leading to susceptibility to health complications and likelihood of premature death. Palliative care may be required at any stage and there may be unpredictable and periodic episodes of care. These include severe cerebral palsy, complex disabilities such as following brain or spinal cord injury, congenital infections (TORCH and Zika virus infection), spina bifida and TB spine.⁴

EXCLUSION CRITERIA

1. Patients with stroke are admitted in a special ward (Spill over wards) in the emergency ward and were not included in this study.
2. Patients on ward admission but who were unstable, those not on bed, those had gone out temporarily for investigations and surgical procedures and those who did not give consent were excluded from the study.
3. Patients admitted in psychiatry wards were excluded

DATA COLLECTION, MANAGEMENT AND ANALYSIS

The information was gathered directly from patients and complemented with that from the patient's case notes. For children, the parents or carers were interviewed. Interviewers made up of fifteen (15) residents in Family Medicine assisted in retrieving information from the participants. Training of research assistants on the various elements of the domains of quality of life and other contents of the proforma for consistency was ensured and pre-test was carried out among the patients admitted in Family Medicine out-patient short-stay observation room. An official request had been made by the principal investigator through The Head, Nursing Services for permission to carry out the evaluation in all wards available for admission of patients. Verbal consent of the participants was received before being interviewed.

DATA MANAGEMENT

Information on demography, diagnosis and when diagnosis was made, chief complaints, stage of disease, domains of palliative care (physical – pain, functional ability, fatigue, strength, poor sleep, appetite, nausea, constipation; psychological – depression, fear, anxiety, enjoyment, cognition, attention, pain distress, happy; social – roles, appearance, relationship, financial constraints, sex function, affection; and spiritual – suffering, hope, religiosity, lost hope in God), category of medical condition (palliative or non-palliative condition), advanced care plan (Will, future treatment plan and possible health outcomes), preparation for home care, death and education about the illness was entered into a data sheet inputted into Epi info version 7.2

DATA ANALYSIS

Descriptive characteristics of the participants were presented as frequencies and percentages for age, sex, pattern of disease, and category of medical conditions (palliative and non-palliative conditions). Chi-squared test was used to

determine association between diagnoses and category of medical condition; stage of disease; advanced care plan, preparation for home care/ death and education on illness. The level of statistical significance was P-value < 0.05.

DATA PRESENTATION AND RESULTS

The total number of hospital beds for admission in the hospital was 950. One hundred and eighty-six (186) of these beds were recruited for the study (from 6 wards selected for the study). Of these, 129 participated in the study. The proportion sampled was 69.4%. Patients excluded were those who were unstable, refused consent, not on bed or gone for investigations.

Age Distribution of Respondents: Fig. 1 shows the age distribution of the respondents. Two-thirds (64.3%) were between the ages of 18 to 59 years, while 15.5% were aged 60 years and above. Only 7.8% of the were less than 5 years.

Distribution of Respondents by Specialty of Managing Medical Doctor.

Figure 2 shows the distribution of the respondents based on the specialty of the managing doctor. About a third of the respondents (30.2%) were managed by Internal Medicine Physicians, 24.0% were in the Surgery non-trauma unit, 20.2% were seen by the Paediatricians, and 17.1% were in the Surgery trauma unit. Only 0.8% of respondents were in the Dental Surgery unit.

Pattern of Ward Admission based on Diagnosis.

Figure 3 shows the pattern of ward admission based on diagnosis. Majority (27.9%) of the respondents admitted into the ward were being managed for cancer. This was followed by trauma (17.8%), infections (10.9%) and hemoglobinopathies (10.9%). Kidney failure, non-cancerous tumour and heart failure accounted for 3.9%, 3.1% and 3.1% diagnoses respectively. Other diagnoses included uterine fibroid (2.3%), stroke (2.3%), and goitre (2.3%). Urinary Obstruction, Tuberculosis of the spine, liver failure, hydrocephalus, congenital anomalies, chronic skin ulcer and AIDS each constituted 1.6% of the diagnoses.

Demographic distribution of the respondents.

Table 1 shows the age distribution of the respondents in the study. The mean age of the respondents was 38.76 ± 20.01 years. About two-thirds (64.3%) of the respondents were between the ages of 18 to 59 years, while 15.5% were aged 60 years and above. Only 7.8% of the respondents were less than 5 years. There were 76 (58.9%) female and 53 (41.1%) male respondents. About 80.6% (104) of the respondents had palliative care conditions, while 19.4% (26) of the respondents had non-palliative care conditions. The Adult respondents were 103 (79.8%).

Table 1
Demographic distribution

AGE (Years)	Frequency (N) 129	Percentage (%)
< 5	10	7.8%
5–17	16	12.4%
18–59	83	64.3%
60 above	20	15.5%
Mean ± SD	38.7674 ± 20.0120	
SEX		
Female	76	58.9%
Male	53	41.1%
Category		
Palliative care condition	104	80.6%
Non-palliative care condition	25	19.4%
Children	26	20.2%
Adult	103	79.8%%
Total	129	100%

Distribution of Ward Admissions.

Table 2 shows the distribution of respondents based on ward admissions. The female medical ward (A3) had 20.9% of the respondents, while the Paediatrics (D2) ward had 17.8%. The male medical (E5), male medical (A2), and male surgical (B3) wards contained 17.1%, 17.1% and 14.7% of the respondents respectively. The female surgical/Gynaecology ward (B1) had 12.4% of the respondents.

Table 2
Distribution of ward admissions

WARD	Frequency (N)	Percentage (%)
A2 Male medical	22	17.1%
A3 Female medical	27	20.9%
B1 Female surgical/Gynaecology	16	12.4%
B3 Male surgical	19	14.7%
D2 Paediatrics	23	17.8%
E5 Male medical	22	17.1%
Total	129	100.0%

Pattern of Ward Admission based on Diagnosis.

Table 3 shows the pattern of ward admission based on diagnosis. Majority (27.9%) of the respondents admitted into the ward were being managed for cancer. This was followed by trauma (17.8%), infections (10.9%) and hemoglobinopathies (10.9%). Kidney failure, non-cancerous tumour and heart failure accounted for 3.9%, 3.1% and 3.1% diagnoses respectively. Other diagnoses include uterine fibroid (2.3%), stroke (2.3%), and goitre (2.3%). Urinary obstruction, tuberculosis of the spine, liver failure, hydrocephalus, congenital anomalies, chronic skin ulcer and AIDS each constitute 1.6% of the diagnoses.

Table 3
Pattern of Ward Admission based on Diagnosis

DIAGNOSIS	Frequency (N)	Percentage (%)
AIDS	2	1.6%
Appendicitis	1	0.8%
Backpain	1	0.8%
Biliary atresia	1	0.8%
Cancer	36	27.9%
Chronic skin ulcer	2	1.6%
Congenital anomaly	2	1.6%
Diabetes	1	0.8%
Goitre	3	2.3%
Heart failure	4	3.1%
Haemoglobinopathy	14	10.9%
Hydrocephalus	2	1.6%
Infections (Non-viral)	14	10.9%
Kidney failure	5	3.9%
Liver failure	2	1.6%
Non-cancerous Tumour	4	3.1%
Seizure	1	0.8%
Spinal cord compression	1	0.8%
Stroke	3	2.3%
Trauma	23	17.8%
Tuberculosis of spine	2	1.6%
Urinary obstruction	2	1.6%
Uterine fibroid	3	2.3%
Total	129	100.0%

Association between Diagnosis and Category of medical condition.

Table 4 shows the association between diagnosis and category of medical condition. There was a statistically significant ($P \leq 0.0001$) association between diagnosis and category of medical condition (Palliative versus non-palliative care condition). The respondents' medical condition was more likely to be a palliative care condition.

Table 4
Association between Diagnosis and Category of medical condition

Variables	Category of medical condition			Statistic	P-value
DIAGNOSIS	Palliative care condition	Non-palliative care condition	Total	$\chi^2 = 129.00$	$< 0.0001^*$
AIDS	2(100.0)	0(0.0)	2(100.0)		
Appendicitis	0(0.0)	1(100.0)	1(100.0)		
Backpain	1(100.0)	0(0.0)	1(100.0)		
Biliary atresia	1(100.0)	0(0.0)	1(100.0)		
Cancer	36(100.0)	0(0.0)	36(100.0)		
Chronic skin ulcer	2(100.0)	0(0.0)	2(100.0)		
Congenital anomaly	2(100.0)	0(0.0)	2(100.0)		
Diabetes	1(100.0)	0(0.0)	1(100.0)		
Goitre	0(0.0)	3(100.0)	3(100.0)		
Heart failure	4(100.0)	0(0.0)	4(100.0)		
Haemoglobinopathy	14(100.0)	0(0.0)	14(100.0)		
Hydrocephalus	2(100.0)	0(0.0)	2(100.0)		
Infections (non-viral)	0(0.0)	14(100.0)	14(100.0)		
Kidney failure	5(100.0)	0(0.0)	5(100.0)		
Liver failure	2(100.0)	0(0.0)	2(100.0)		
Non-cancerous Tumour	0(0.0)	4(100.0)	4(100.0)		
Seizure	1(100.0)	0(0.0)	1(100.0)		
Spinal cord compression	1(100.0)	0(0.0)	1(100.0)		
Stroke	3(100.0)	0(0.0)	3(100.0)		
Trauma	23(100.0)	0(0.0)	23(100.0)		
Tuberculosis of spine	2(100.0)	0(0.0)	2(100.0)		
Urinary obstruction	2(100.0)	0(0.0)	2(100.0)		
Uterine fibroid	0(0.0)	3(100.0)	3(100.0)		
Total	104(80.6)	25(19.4)	129(100.0)		

Summary Distribution of Physical Palliative Domain.

Table 5A shows the summary distribution of physical palliative domain. Pain constitutes 94.6% of all physical palliative care features, while functional ability constituted 83.7%. About two-thirds of the respondents (65.1%) had fatigue, while 58.9% had reduced strength as part of their features. Poor sleep, poor appetite, and nausea constituted 55.0%, 20.9%, and 19.4% respectively. The least represented palliative care feature was constipation (7.0%).

Table 5
A: Summary distribution of physical palliative domain

PHYSICAL PALLIATIVE CARE FEATURES	Frequency (N)	Percentage (%)
*Pain	122	94.6%
*Functional ability	108	83.7%
*Fatigue	84	65.1%
*Strength	76	58.9%
*Poor sleep	71	55.0%
*Appetite	27	20.9%
*Nausea	25	19.4%
*Constipation	9	7.0%
· Multiple responses		

Summary Distribution of Psychological Palliative Domain

Table 5B shows the summary distribution of psychological palliative domain. Depression was the commonest (81.4%) psychological domain, followed by fear (78.3%). Less than half (42.6%) of the respondents had anxiety. Other psychological domains affected were enjoyment (6.2%), cognition (5.4%), attention (3.9%), pain distress (3.1%) and happiness (1.6%). About 7.0% of respondents had no psychological feature affectation reported.

Table 5
B: Summary distribution of psychological palliative domain

PSYCHOLOGICAL PALLIATIVE CARE FEATURES	Frequency (N)	Percentage (%)
*Depression	105	81.4%
*Fear	101	78.3%
*Anxiety	55	42.6%
*Enjoyment	8	6.2%
*Cognition	7	5.4%
*Attention	5	3.9%
*Pain distress	4	3.1%
*Happy	2	1.6%
*Nil	9	7.0%
· Multiple responses		

Summary Distribution of Social Palliative Domain.

Table 5C shows the summary distribution of social palliative domain. Roles, appearance and relationships made up 63.6%, 63.6% and 62.0% respectively. Financial constraint constituted about half (51.9%), while sexual function made up 20.2%.

Table 5
C: Summary distribution of social palliative domain

SOCIAL PALLIATIVE CARE FEATURES	Frequency (N)	Percentage (%)
*Roles	82	63.6%
*Appearance	82	63.6%
*Relationship	80	62.0%
*Financial constraint	67	51.9%
*Sex function	26	20.2%
*Affection	4	3.1%
*Nil	8	6.2%
· Multiple responses		

Summary Distribution of Spiritual Palliative Domain.

Table 5D shows the summary distribution of spiritual palliative domain. This consists of suffering (88.4%), Hope (80.6%), Religiously in God (41.1%) and Loss of hope in God (0.8%).

Table 5
D: Summary distribution of spiritual palliative domain

SPIRITUAL PALLIATIVE CARE FEATURES	Frequency (N)	Percentage (%)
*Suffering	114	88.4%
*Hope	104	80.6%
*Religiously in God	53	41.1%
*Lost hope in God	1	0.8%
*Nil	9	7.0%
· Multiple responses		

Table 6 shows the patients' admission characteristics. Concerning the caregivers, 108 (83.7%) respondents had family members as their caregiver, while 3.1% had others (paid caregivers, neighbours) as caregivers. About one-eighth (13.2%) of the participants had no care-giver. About two-thirds of the respondents (65.9%) had no advanced care plan, while about a third of them (34.1%) had an advanced care plan. There was no preparation for home care or death in 72.1% of the respondents, while 20.2% had made preparation for death, and 7.8% had made preparation for home care.

More than two-thirds (70.5%) of the respondents had education about their illness, while 29.5% had no education about their illness. More than two-thirds of the respondents (68.2%) were in the advanced stage of their disease,

while 31.8% were in the early stage of their disease.

Table 6
Patients' Admission characteristics

CAREGIVER	Frequency (N)	Percentage (%)
Family	108	83.7%
Others (Paid caregiver, neighbours, etc)	4	3.1%
Nil	17	13.2%
Total	129	100.0%
ADVANCED CARE PLAN	Frequency (N)	Percentage (%)
No	85	65.9%
Yes	44	34.1%
Total	129	100.0%
PREPARATION FOR HOME CARE / DEATH	Frequency (N)	Percentage (%)
Death	26	20.2%
Home care	10	7.8%
Nil	93	72.1%
Total	129	100.0%
EDUCATION ABOUT ILLNESS	Frequency (N)	Percentage (%)
No	38	29.5%
Yes	91	70.5%
Total	129	100.0%
STAGE OF DISEASE	Frequency (N)	Percentage (%)
Advanced	88	68.2%
Early	41	31.8%
Total	129	100.0%

Association between Specialty and Advance Directives

Table 7 shows the association between Specialty and Advance directives. This association was statistically significant ($P = 0.0038$). Participants attending the surgery non-trauma unit were more likely (51.6%) to have advance directives, followed by surgery trauma unit (50.0%), and paediatrics unit (38.5%) respondents. The prevalence of advance directives among obstetrics and gynaecology speciality and Medicine were 30.0% and 10.3% respectively.

Table 7
Association between Specialty and Advance directives

Variables	Advance directives			Statistic	P-value
Specialty	No	Yes	Total	$\chi^2 = 17.38$	0.0038*
Surgery Non-Trauma	15(48.4)	16(51.6)	31(100.0)		
Surgery Trauma	11(50.0)	11(50.0)	22(100.0)		
Medicine	35(89.7)	4(10.3)	39(100.0)		
Obstetrics and Gynaecology	7(70.0)	3(30.0)	10(100.0)		
Dental Surgery	1(100.0)	0(0.0)	1(100.0)		
Paediatric	16(61.5)	10(38.5)	26(100.0)		
Total	85(65.9)	44(34.1)	129(100.0)		
· Significant p-value < 0.05					

Association between Demography and Caregiver

Table 8 shows the association between Demography and Caregiver. Respondents between the ages of 5–17 were more likely to have family members as their caregiver (100%), followed by those less than five years (90%), and those between the ages of 18–59 (81.9%). However, there was no statistically significant association between the age categories and caregiver ($P = 0.1659$).

There was a statistically significant association between gender and caregiver. Males were more likely (96.2%) to have family members as caregivers than females. However, females were more likely to have other caregivers (3.9%) such as neighbours and paid caregiver, compared to males (1.9%). The proportion of participants that had no caregiver at all were more common among the females (21.1%) than males (1.9%)

Table 8
Association between Demography and Caregiver

Variables	Caregiver				Statistic	P-value
AGE (Years)	Family	Others	Nil	Total	$\chi^2 = 9.1395$	0.1659
< 5	9(90.0)	1(10.0)	0(0.0)	10(100		
5–17	16(100.0)	0(0.0)	0(0.0)	16(100		
18–59	68(81.9)	3(3.6)	12(14.5)	83(100		
60 above	15(75.0)	0(0.0)	5(25.0)	20(100		
Total	108(83.7)	4(3.1)	17(13.2)	129		
Mean $\pm$ SD	37.4713 $\pm$ 20.3559	34.7750 $\pm$ 23.3653	47.9412 $\pm$ 15.0228		F = 2.13	0.1232
Sex						
Female	57(75.0)	3(3.9)	16(21.1)	76(100	$\chi^2 = 10.8$	0.0045*
Male	51(96.2)	1(1.9)	1(1.9)	53(100		
Total	108(83.7)	4(3.1)	17(13.2)	129(100		
· Significant p-value < 0.05						

Association between Patients' Admission Characteristics and Caregiver

Table 9 shows the association between patients' admission characteristics and caregiver. There was no statistically significant association between having an advanced care plan and caregiver status of the respondent ($P = 0.09$). Also, participants who prepared for home care and death were more likely to be those who had family care-givers (100% and 92.3% respectively). However, there was no statistically significant association between preparation for homecare/death and caregiver status.

Participants who had education about the illness were more likely to have a family member as a caregiver {92.3%}, compared to those who had no education about the illness (63.2%). This was a statistically significant association ($P < 0.05$). There was no significant association between the stage of the illness (early or advanced) and the caregiver status.

Table 9
Association between Patients Admission characteristics and Caregiver

Variables	Caregiver				Statistic	P-value
ADVANCE CARE PLAN	Family	Others	Nil	Total	$\chi^2 = 4.64$	0.0984
No	67(78.8)	3(3.5)	15(17.6)	85(100.0)		
Yes	41(93.2)	1(2.3)	2(4.5)	44(100.0)		
Total	108(83.7)	4(3.1)	17(13.2)	129(100.0)		
PREPARATION FOR HOME CARE / DEATH	Family	Others	Nil	Total	$\chi^2 = 4.81$	0.3072
Death	24(92.3)	0(0.0)	2(7.7)	26(100.0)		
Home care	10(100.0)	0(0.0)	0(0.0)	10(100.0)		
Nil	74(79.6)	4(4.3)	15(16.1)	93(100.0)		
Total	108(83.7)	4(3.1)	17(13.2)	129(100.0)		
EDUCATION ABOUT ILLNESS	Family	Others	Nil	Total	$\chi^2 = 17.37$	0.0002*
No	24(63.2)	2(5.3)	12(31.6)	38(100.0)		
Yes	84(92.3)	2(2.2)	5(5.5)	91(100.0)		
Total	108(83.7)	4(3.1)	17(13.2)	129(100.0)		
STAGE	Family	Others	Nil	Total	$\chi^2 = 1.54$	0.4642
Advance	76(86.4)	2(2.3)	10(11.4)	88(100.0)		
Early	32(78.0)	2(4.9)	7(17.1)	41(100.0)		
Total	108(83.7)	4(3.1)	17(13.2)	129(100.0)		
· Significant p-value < 0.05						

Association between Demography and Category of medical condition

Table 10 shows the association between Demography and Category of medical condition. There was a statistically significant association between age of respondents and category of medical condition (palliative versus non-palliative care). Participants between the ages of 18–59 were more likely to have a palliative care condition (88.0%), followed by those less than 5 years of age (80.0%) and those greater than 60 years (80.0%). The mean age of respondents who had palliative care condition was 39.94 ± 19.01 . Adults were more likely to have palliative care conditions (86.4%) compared to children (57.7%), and this was significantly associated ($P < 0.05$)

Respondent who were female were more likely to have a palliative care condition (84.2%) compared to those who were male (75.5%). However, there was no significant association between gender and category of medical condition. ($P = 0.1565$)

Table 10: Association between Demography and Category of medical condition

Variables	Category of medical condition				Statistic	P-value
Age	Palliative care condition	Non-palliative care condition	Total		$\chi^2 = 16.78$	0.0008*
<5	8(80.0)	2(20.0)	10(100.0)			
5-17	7(43.8)	9(56.3)	16(100.0)			
18-59	73(88.0)	10(12.0)	83(100.0)			
60 above	16(80.0)	4(20.0)	20(100.0)			
Total	104(80.6)	25(19.4)	129(100.0)			
Mean $\pm$ SD	39.9404 $\pm$ 19.0124	33.8880 $\pm$ 23.5332			T = 1.36	0.1755
	Children	15(57.7)	11(42.3)	26(100.0)	$\chi^2 = 0.001$	0.0021*
	Adult	89(86.4)	14(13.6)	103(100.0)		
	Female	64(84.2)	12(15.8)	76(100.0)	$\chi^2 = 0.22$	0.1565
	Male	40(75.5)	13(24.5)	53(100.0)		
	Total	104(80.6)	25(19.4)	129(100.0)		

Significant p-value <0.05

DISCUSSION

The research was carried out among patients admitted into the wards at Lagos University Teaching Hospital, Idi-Araba. It was aimed at examining the pattern of admissions and needs assessment for palliative services among these patients in order to justify the need for the establishment of palliative care services in Lagos University Teaching Hospital Idi-Araba Lagos Nigeria

Sociodemographic Characteristics of Respondents

A total of 129 patients were interviewed for this study. Majority of the respondents were aged between 18–59 years, representing 83% of them. This was similar to the findings made by Ndiok and Ncama where this same age group represented 85.7% of their participants.⁹ The similarity could be attributed to the settings where recruiting of the respondents was done. This study, and their study were both conducted in Nigeria, and in teaching hospitals. Their mean age was 44.8 ± 14.5 years while this study's finding was 38.77 ± 20.0 years.

This finding was, however, contrary to Robinson *et al*'s study (in New Zealand) who had majority of their participants being in the age range of 60–79 years (representing 54.3%).¹⁰ This may be because of the difference in their recruitment method. Although Robinson *et al*'s study was also a cross-sectional study, the least age for the recruited participants was 18 years and they all had the ability to understand and speak English.

In this study, more females were respondents, making the sex distribution to be 59% females and 41% males, and a ratio of 1.44:1. Most studies reviewed also had a similar pattern of more female respondents. Ndiok and Ncama, Robinson *et al*, and Agbodande *et al* (in Benin Republic) all had higher preponderance of females as 61.1%, 50.9% and

61% respectively in their studies as well.^{9,10,11} However, the study finding of Olden *et al* in New York, in relation to sex distribution was dissimilar.¹² The female and male distribution was 46.44% and 53.56% respectively.

Pattern of Diseases Among Admitted Patients

In this study, several diseases were diagnosed in the respondents. For better collation and analysis, these diseases were further divided into different categories.

Pattern of Diseases Based on ICPC-2

The International Classification of Primary Care (ICPC) is the most widely used international classification for systematically capturing and ordering clinical information in Primary Care.¹³ It was developed and updated by the World Organization of Family Doctors' (WONCA) International Classification Committee (WICC), and the most recent version is the ICPC-2 which was revised in 2015.¹³ The ICPC is divided into 17 chapters by body systems representing the localisation of the problem and/or disease. This makes it easy to use for healthcare providers.¹³

Based on ICPC-2 classification, in this study, Musculoskeletal-related diseases were observed to have the highest prevalence of 20.2%, and this was followed by the Blood, blood-forming organs and Immune mechanism-related diseases at 17.1%. The least observed were the Male genital-related and Psychologically-related diseases which were both 0.8% of the identified diseases in the respondents.

Contrary to the findings of this study, Ayankogbe *et al* reported in their study that General and unspecified illnesses had the highest prevalence of 23.1%, this was followed by Pregnancy, child-bearing and family planning-related illnesses at 13.9%.¹⁴ Their least recorded pattern of illnesses were CVDs (1.7%) and Psychological diseases (1.4%).¹⁴ Although this study, and Ayankogbe *et al*'s study were conducted in the same geographical area and with similar participants and sampling technique, the difference in the reported findings may be because Ayankogbe *et al* had private general/family practice clinics as their study sites as compared with this study which had a teaching hospital as its study site. Of note, however, is the similarity with this study, where psychological diseases were the least occurring in the geographical area.

Also, Gataa *et al* reported differences in the pattern of illnesses with highest prevalence as compared to this study.¹⁵ They reported Respiratory and Digestive illnesses to be of the highest prevalence at 43% and 10% respectively.¹⁵ Surprisingly, however, the diseases with least prevalence, as reported by Gataa and colleagues were similar to those reported in this study. They were Male genital-related and Psychologically-related diseases which were both 0.1% and 0.85% respectively. The differences in the pattern of illnesses with highest prevalence as compared with this study may be because of the sampling technique employed and the lifestyle of the recruited participants.

Pattern of Diseases Based on Clinical Assessment/ Provisional Diagnosis

In this study, Cancer was recorded as the highest occurring disease in the participants on admission, the prevalence was 27.9%; and this was followed by Trauma (having a prevalence of 17.8%); the least-occurring diseases were Appendicitis, Backpain, Biliary atresia, Diabetes, Seizures and Spinal cord compression, all had similar prevalence of 0.8% among the participants on admission.

In a comparative study by Nicholson *et al*, Cancer was also reported to be the highest occurring disease at 72% when the standard specialist care of palliative care was applied to their participants.¹⁶ The Bromley Care Coordination (BCC) which was being compared with, however, had 16% prevalence of cancerous diseases. The most prevalent

disease in the standard care (cancer) as compared to this study may be because the study was conducted among individuals who needed palliative care, moreover, they were elderly individuals. The BCC findings, however, had lower cancerous cases possibly because it was a newly introduced innovative home care service, primarily for older people with palliative care needs who do not meet the criteria for referral to specialist palliative care.¹⁶

Also, Robinson *et al* and Agbodande *et al* reported higher cancer prevalence compared to non-cancer diseases in their studies, as 81.9% and 89.2% respectively.^{11,18} Robinson *et al*'s study was conducted in New Zealand, a country in Oceania continent among individuals aged 18 years and above; while Agbodande *et al*'s was in Benin, a West African country, just like the setting of this study, but with age distribution of 16–82 years, unlike this study with no age restriction in the recruited inpatient-participants.

In contrast to the findings of this study, Dinçer *et al* reported in their study that Cancer was of the third highest prevalence (23.4%), after Neurologic diseases (42.3%) and Chronic systemic diseases (41.4%).¹⁷ The least occurring disease that was reported by them was Infection (10.8%). The difference in their findings, and this study's may be because of the difference in the settings of both studies, moreover, Dinçer *et al* recruited only geriatric participants, unlike this study which also recruited non-geriatric patients.

Pattern of Diseases Based on Category of Medical Condition.

In this study, most of the participants were eligible for palliative care. They represented 80.6% of the total participants. This is despite the fact that the recruited patients were general in-patients with numerous and diverse disease conditions. This further confirmed that palliative care arrangement in every tertiary health care facility cannot be over-emphasized. All the reviewed research results also buttressed this point.

Distribution of Palliative Care Domains of Patients

In this study, the four major domains of palliative care which include physical, psychological, social and spiritual domains were considered and several features were also considered under each major domain. Physical domain was the most occurring, with the highest feature being Pain and representing 94.6% (the least occurring feature was Constipation-7%), the next occurring domain was Spiritual (the highest feature was Suffering-88.4% and least feature was lost hope in God-0.8%). Social palliative domain had the least occurring domain at 63.6%. Agbondade *et al* reported findings that were similar to that of this study. The highest reported feature in their study was Pain which was seen in all their participants (100%), they also observed that the next occurring domain was Spiritual which represented 90.2% of the participants and the next occurring domain was Psychosocial which represented 73%.¹¹ The similarity in the distribution of the domains may be related to the settings where the researches were conducted, both were done in West Africa. When compared to this study, Anderson *et al*'s findings were dissimilar. The highest domain of care that was recorded in their study was the Social domain which represented 37% of the results, this was then followed by other domains of care that trailed one another: Physical (15%), Psychological and Psychiatric (15%), and Spiritual, religious and existential domains (14%).¹⁸ The difference in reported findings with this study and Anderson *et al*'s may be because of the setting the research was conducted, it was conducted in the US, and the participants were not only patients on admission, like in this study, but they included patients, family/care givers and professionals.

Admission Characteristics among Patients

In this study, the patient admission characteristics put into consideration were: the caregiver advanced directive, preparation for home care/death, education about illness and stage of the disease. Most of the caregivers rendering

care to participants in this study were the family members (83.7%), there was no previous advanced directive by most of the participants (65.9%), and most of them were not prepared for death/home care (72.1%). Although more than two-thirds of the participants knew about their illness (70.5%), the stage of the disease was mostly advanced (68.2%) in them. Several reviewed works had different admission characteristics of patients that they reported on. In this study, a sizeable number of the caregivers (83.7%) were the patients' family members (relatives). This was the discovery made in other reviewed studies as the caregivers who were all relatives were spouse, children, sibling(s), daughter/son-in-law or parent(s). In the studies by Mercandes *et al*, Oğuz *et al* and Monsomboon *et al*, the family represented 100% of the caregivers.^{19,20,21} This confirmed the importance of the family in the role of care of an individual. In this study, most of the participants did not make provision for advanced directive, they constituted 65.9% of the participants. This was, however, contrary to the findings of Monsomboon *et al* who reported that 60.4% of their palliative care patients made previous advanced care plan, 4.9% also had living will, while 18.1% also had documented medical records.²¹ Monsomboon *et al*'s study was done in Thailand, and this may be the reason for the difference in the findings with this study. In this study, enquiries were made to find out the preparation of the participants for home care or death, majority of the participants (72.1%) did not have such plans. This finding was similar to that of Monsomboon and colleagues. They found out that 72.5% of the participants in their study had no plans for where they prefer to die, however, 15.9% preferred to die at home while 9.3% preferred the hospital.²¹ They also considered the caregivers, making enquiries about their preferred place of death for the patients they were taking care of. Majority of the participants had no plans (37.4%), the hospital was preferred in 36.3%, and 23.1% preferred the home.²¹ The findings may suggest that no one plans or prepares to die because no one wants to die. In this study, most of the participants were aware of their illness (70.5%). This was similar to the findings made by Mercadante *et al* who discovered that among the patients in their study on planned admission, 51.2% were completely aware of their disease while 39.0% were partially aware.¹⁹ However, of the patients on unplanned admission, 57.4% were completely aware, while 35.3% only had partial awareness.¹⁹ They went further to access the caregivers' awareness status. The caregivers of patients on planned admission who were completely aware of their disease represented 86.2%, while 11.0% were only partially aware; the caregivers of patients of unplanned admission represented 83.8% while only 13.2% were partially aware of the disease condition of the patients they were taking care of.¹⁹ Only a minute number of caregivers (2.8% and 2.9% respectively) in both categories of patients had no awareness of the disease condition of the patients under their care. Monsomboon *et al* while studying the characteristics and factors associated with mortality in palliative patients also accessed the caregivers' understanding about the palliative status of the patients they were taking care of.²¹ Majority of the participants (79.7%) were aware of the palliative status of the disease condition of the patients, while only 13.2% were not.²¹ Contrary to previous reported findings on patient's awareness of the disease condition, Oğuz *et al* reported that 81.3% of the advanced cancer patients admitted to the palliative care unit had absent information about their disease, only 18.7% indicated presence of knowledge.²⁰ Mercadante *et al* while describing the characteristics of patients who had unplanned admission in comparison to those with planned admission to an acute palliative care unit reported that most of the patients had their diseases in the advanced (metastatic) stage.¹⁹ The patients with unplanned admission had 82.4% of them with metastatic disease while 74.0% of those with planned admission had metastatic diseases.¹⁹ These findings look similar to that from this study where majority of the participants (68.2%) on admission had advanced stage of disease. This was also similar to the findings made by Oğuz *et al* who found out in their study in India, that 96.4% of the patients admitted to the palliative care unit already had metastasis.²⁰ These findings may corroborate the fact that most patients present to health facilities in advanced stage of their diseases.

CONCLUSION

Palliative care service was demonstrated as very important management option in the study, with majority of the participants needing palliative care. Most of the participants were in the economic productive age of 18–59 years, there were also more females and the most predominant health condition for admission into the wards was cancer. The most common palliative domain was physical, and pain was the commonest feature. Most of the patients were taken care of by their family members and they had knowledge about their illness. It was however, surprising that majority had no advanced directive or preparation for home care/death despite having advanced stages of disease.

STRENGTH

1. This is the first Nigerian study to evaluate the admission pattern and needs assessment for Palliative care services among in-patients admitted in a tertiary healthcare facility
2. This study also employed the use of Together for Short Lives (TfSL) tool to categorize the participants medical diagnoses into Palliative and Non-Palliative conditions
3. The study site has the largest cancer centre in Nigeria

LIMITATIONS

1. Some blocks of wards were under renovation and limited the number of ward admissions
2. Recruitment was done within two days limiting the chance of being recruited
3. The design was cross-section and disallowed follow up of patients' experience

Declarations

ETHICAL APPROVAL AND CONSENT TO PARTICIPATE

We confirm that all the study was carried out in accordance with relevant guidelines and regulations (Declaration of Helsinki). Ethical approval was obtained from the Health Research and Ethics Committee (HREC) from Lagos University Teaching Hospital Idi-Araba (ADM/DSCST/HREC/APP/5246). A verbal Informed consent, with all risks clearly stated was developed and attached to the protocol. For participants below 18 years of age, Informed consent was received from the parent / guardian staying with the participants. All participants were given the right to withdraw from the study at any time without any affectation of their continued care.

CONSENT FOR PUBLICATION

Not Applicable

COMPETING INTEREST

Authors declare that there is no competing interest

AVAILABILITY OF DATA AND MATERIALS

Data and other relevant materials used for the study will be made available on request

FUNDING

This study did not receive any specific funding

AUTHORS' CONTRIBUTIONS

Akodu BA: conceived, designed, analyzed, interpreted and drafted the manuscript

Olokodana-Adesalu A O: wrote the discussion and reviewed the manuscript

Ojikutu M S: data acquisition and wrote the result comment and reviewed the manuscript

Ogbenna A: reviewed the manuscript for intellectual contribution

Agunbiade HT: data acquisition and reviewed the manuscript for intellectual contribution

Nwosa Nwando: data acquisition and reviewed the manuscript for intellectual contribution

Ashti-Doobay P: conceived and reviewed the manuscript for intellectual contribution

Matthew TC: reviewed the manuscript for intellectual contribution

Denise LD: reviewed the manuscript for intellectual contribution

Charlesnika TE: reviewed the manuscript for intellectual contribution

Adeboye O: reviewed the manuscript for intellectual contribution

Joshua MH: conceived and reviewed the manuscript for intellectual contribution

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Figures

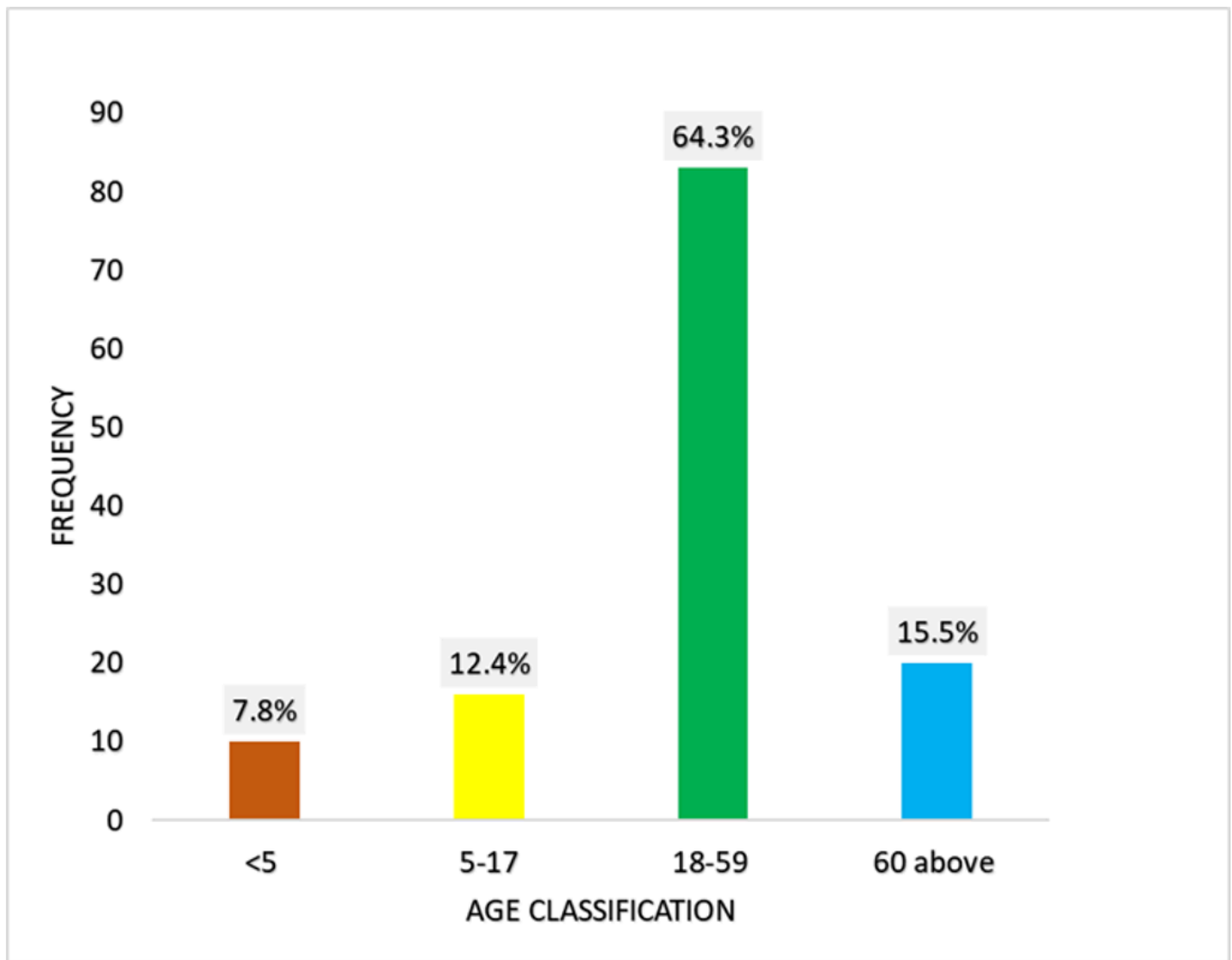


Figure 1

Age distribution of respondents

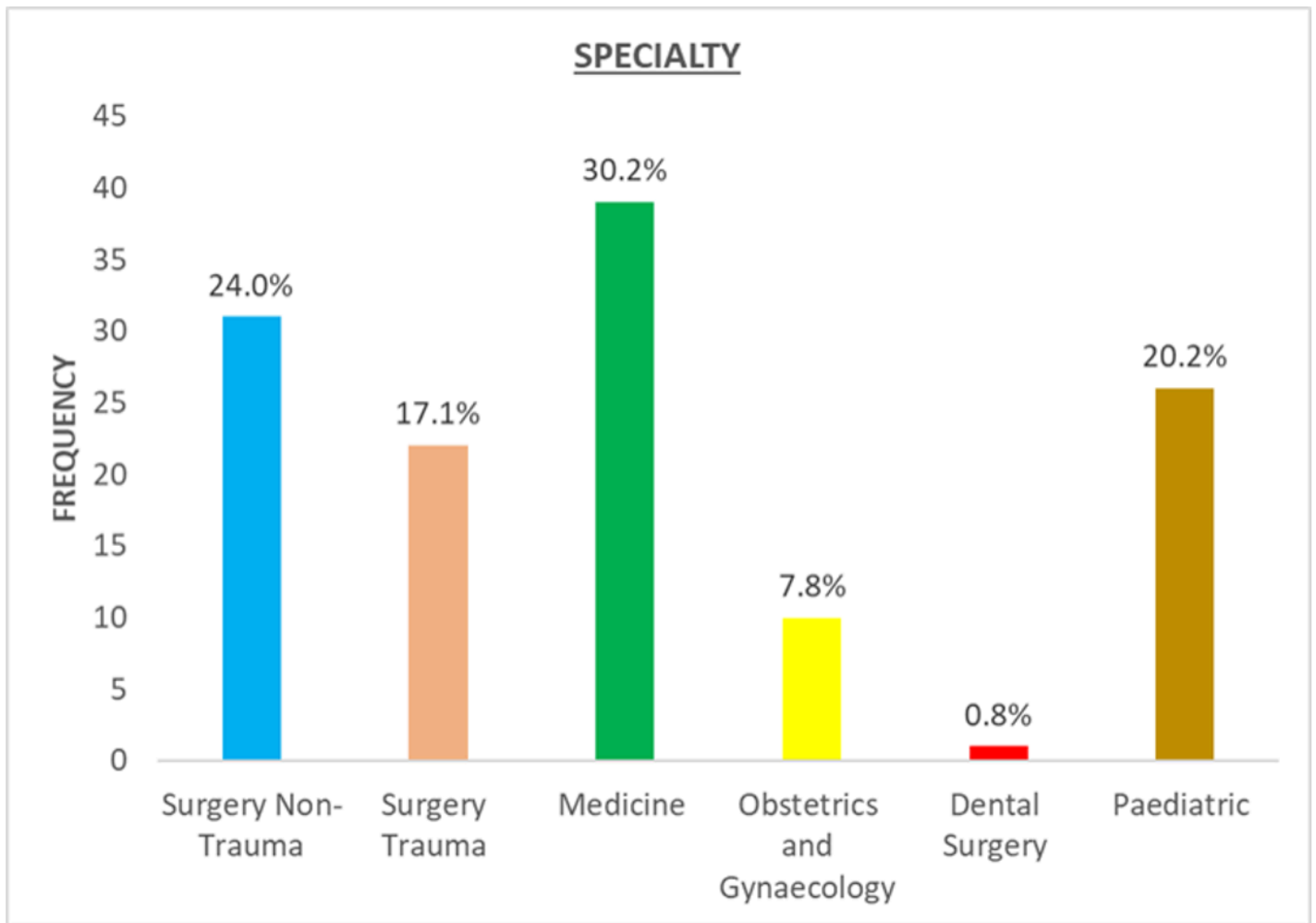


Figure 2

Distribution of respondents by specialty of managing medical doctor

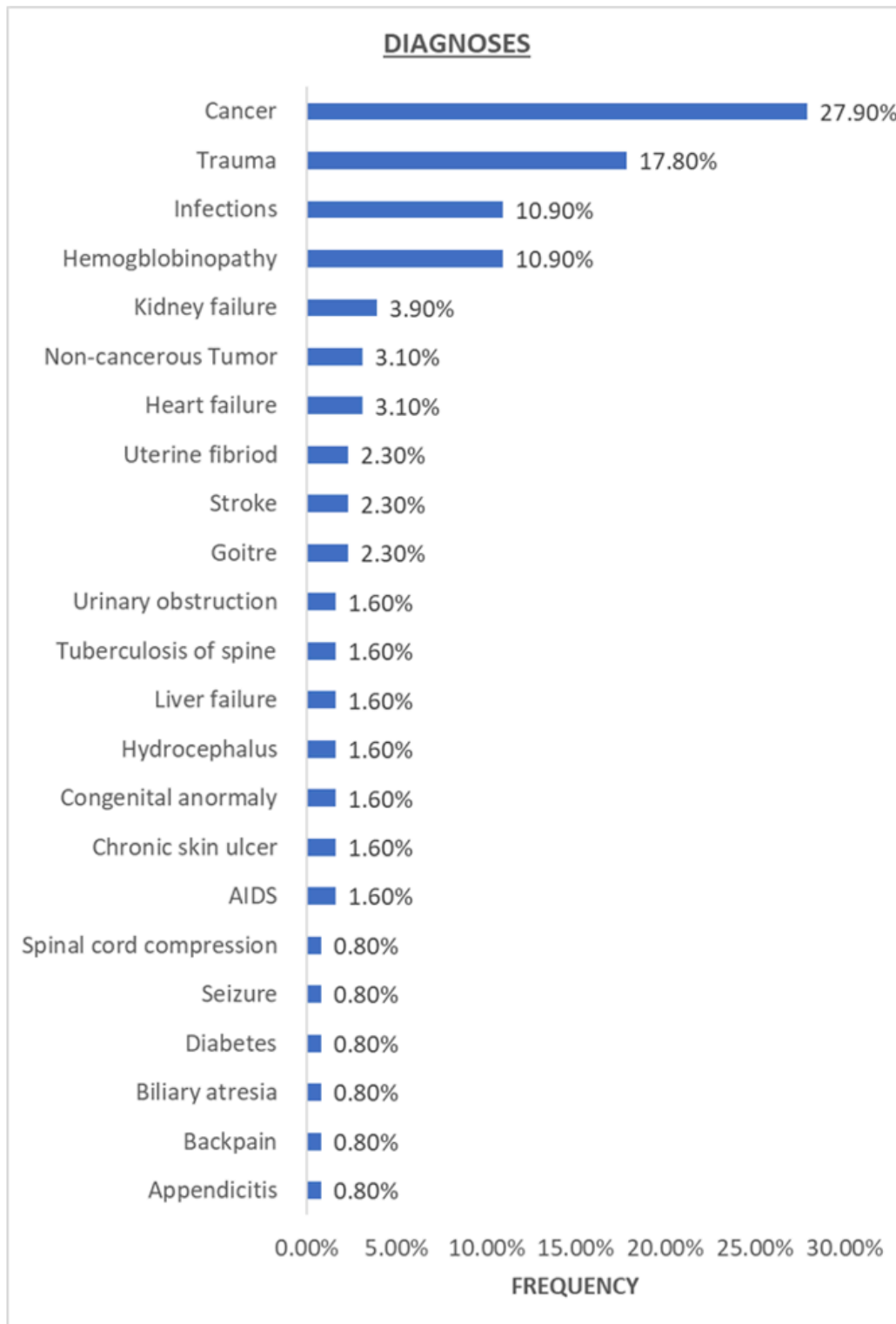


Figure 3

Pattern of ward admissions based on diagnoses